

Internet of Things: Intruder Detection Robot



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Introduction

Intruder detection system that integrates various sensors and technologies to enhance security.

- Send update to user when intruder detected
- Detects unwanted objects
- Turns on alerts when intrusion occurs
- Protects expensive items
- Guards restricted areas





Literature Review

Physical Intrusion Detection Systems with Sensor Technology

Sensor Technologies:

- Utilizes motion, infrared, and vibration sensors to detect unauthorized access.

User Notification:

- Sends real-time alerts via SMS, email, or mobile apps upon intrusion detection.

Alert Activation:

- Triggers auditory alarms and visual signals; integrates with security control centers for rapid response.

Smart Integration:

- Enhances system capabilities through integration with smart home technology and IoT.
- IoT-based smart environments (cities, homes) utilize layered security architectures from physical devices to management systems, emphasizing the need for robust security measures.



Literature Review

Advanced Techniques:

- Incorporation of **machine learning** and **bio-inspired algorithms** enhances the capability to detect subtle and complex intrusion patterns.

IoT Security Challenges:

- Integration complexities with existing infrastructure.
- High data volume management from numerous IoT devices.
- Ensuring real-time, accurate threat detection.

Regulatory Compliance:

- Compliance with data protection regulations (GDPR, HIPAA) is critical, necessitating privacy and data integrity alongside intrusion protection.



Market Research

Statistics on physical intrusions and break-ins in Singapore and worldwide:

Singapore

In 2023, there were 290 reported cases of housebreaking and theft, marking a 17.9% increase from the previous year. This rise was primarily due to an increase in residential unit break-ins

Global

Globally, there were approximately 522,426 burglaries in the United States alone in 2020, with 62% being residential break-ins

Trends

There is a rising trend in adopting smart home security systems worldwide, driven by technological advancements and the need for improved security

Market Research



Growth Projections

The global IoT market is growing rapidly - USD 1.6 trillion by 2025



Technological Trends

Increasing use of AI and machine learning to enhance system responsiveness and threat adaptation.



IoT and Cloud Integration

Combining cloud computing and IoT enhances data management and security in smart environments, supporting real-time intrusion detection and response.

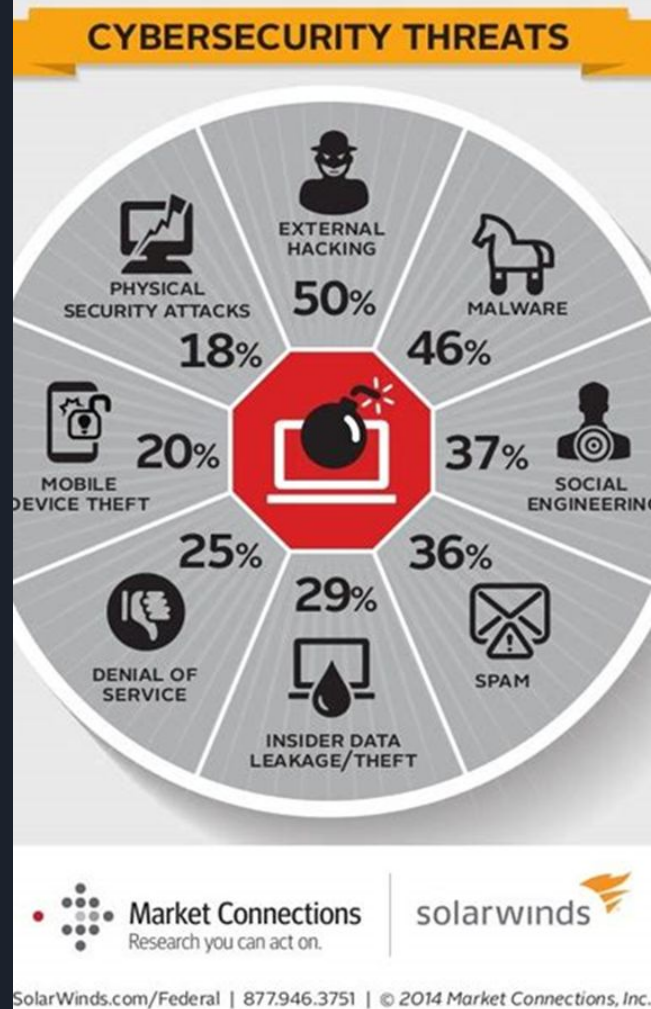
Demand Drivers and Market Challenges

Demand Drivers

- Security concerns
- Smart home adoption
- Technological advancement

Market Challenges

- Privacy Concerns (Data Security & Surveillance Misuse)
- Reliability and False Alarms (Accuracy, Environmental Factors)

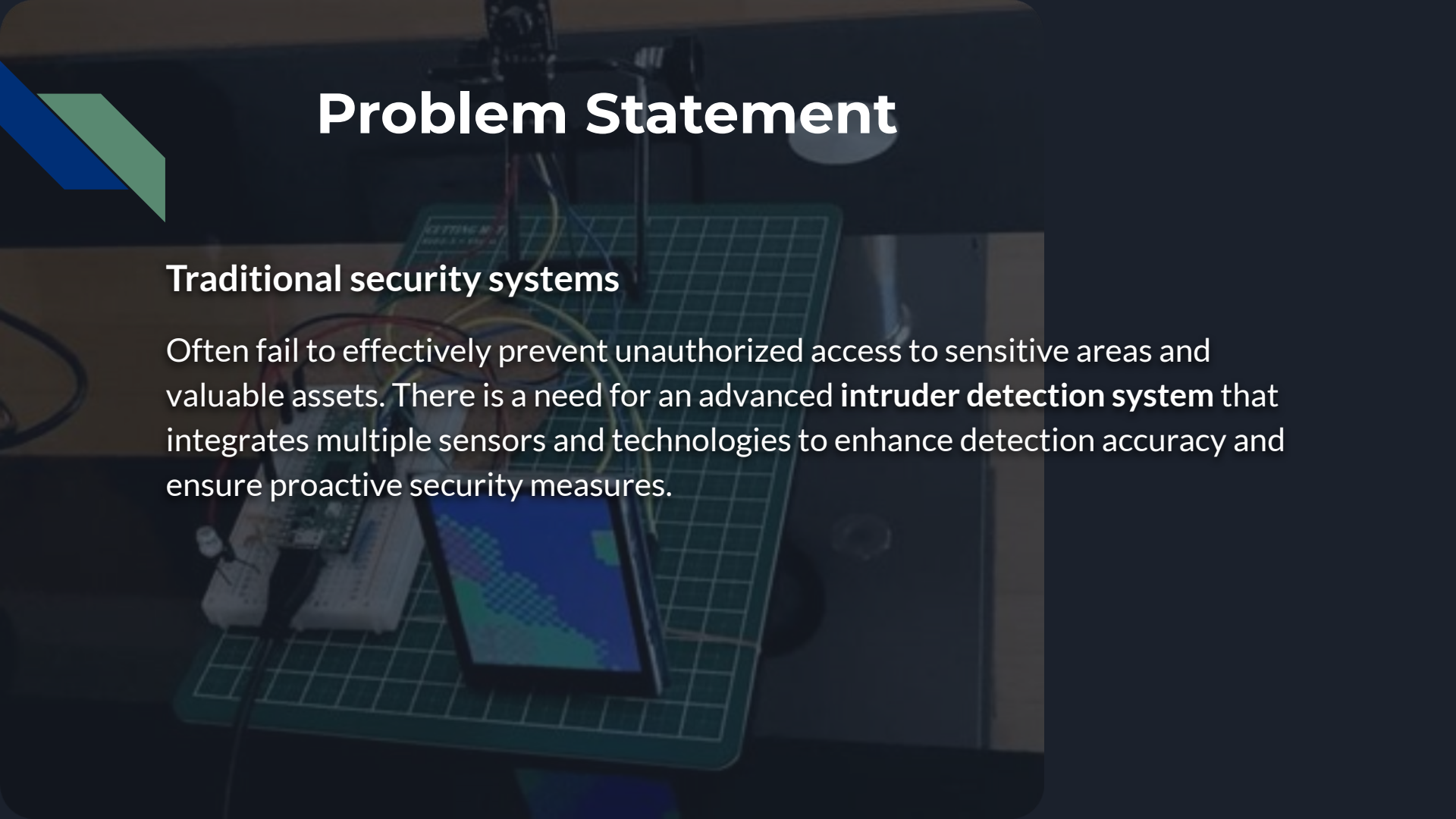




Problem Statement

Traditional security systems

Often fail to effectively prevent unauthorized access to sensitive areas and valuable assets. There is a need for an advanced **intruder detection system** that integrates multiple sensors and technologies to enhance detection accuracy and ensure proactive security measures.



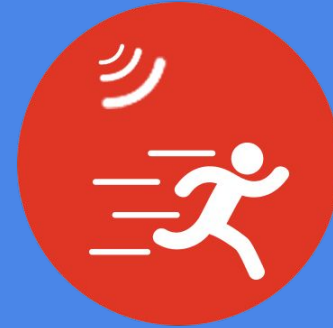
Intruder Detection Innovation : Objectives

Monitoring and surveillance system



- ❑ Use motion detection
- ❑ Monitor specific surrounding 24/7

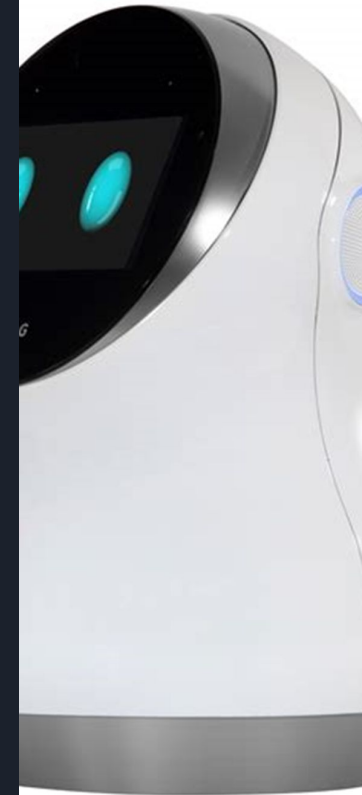
Real-time detection and alert systems



- ❑ Detecting, assessing, and responding to potential security threats with minimal human oversight

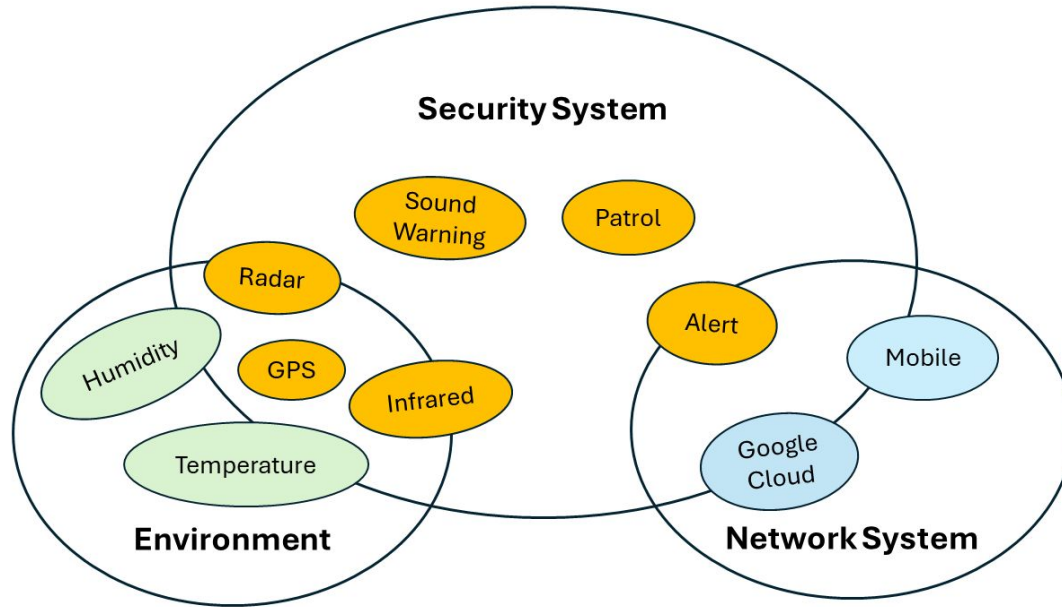
IoT Characteristics for Intruder Detection

Trustworthiness	Architectural	Functional
● Confidentiality It is crucial that all data it captures, processes, and transmits is protected from unauthorized access and disclosure. ● Safety The robot operates in environments possibly shared with humans, ensuring it does not pose a threat to people or property is crucial.	● Network Connectivity The intruder detection robot must be capable of connecting to a network to communicate alerts, receive updates, and coordinate with other security systems or devices.	● Real-time capability Immediate action can be the difference between preventing a crime and merely recording it. Process information and respond in real-time is essential for a security-related device. ● Context-awareness To be aware of its environment allows the robot to make intelligent decisions based on the context of the situation. For example, recognizing whether an area is supposed to be vacant or occupied can influence alert decisions.

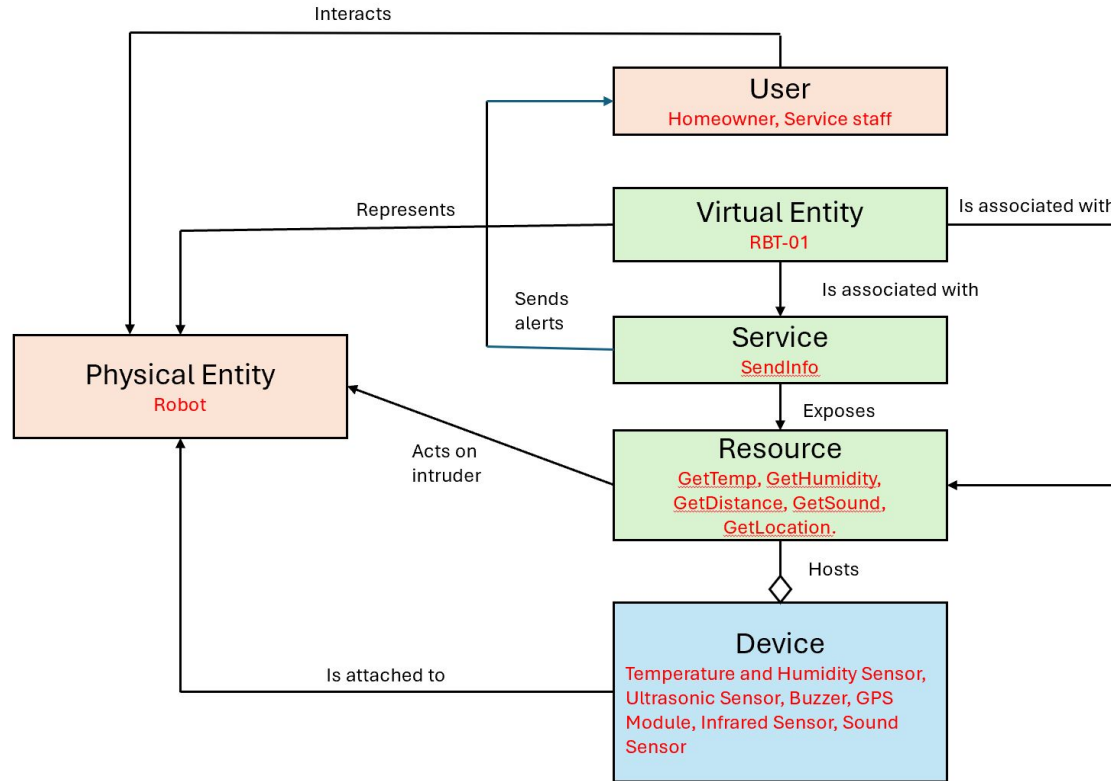


Context Model

IoT Environment for Intruder Detection Robot



Entity Model





Functional Viewpoints

Physical Interaction:

Sense function that reads from the distance sensor

Data function:

Handling sensor data, processing it for decision-making, and triggering alarms or notifications.

Management function:

Overseeing system performance, updating software, managing user access, and ensuring system security.

Communication function:

Device Layer: (DHT11/Digital Temperature and Humidity sensor) (HC-SR04/ Ultrasonic Sensor) (Buzzer) (GY-NEO 6M V2/ GPS module) (Sharp GP2Y0A02YKOF/ Analog Infrared sensor) (Sound Sensor)

Connectivity Layer: Wi-Fi 6

Data Processing Layer: (Google Cloud IoT Platform)

Application Layer: (Google Cloud IoT Platform) (HTTP)



Implementation Viewpoints

Transducer Capabilities

Sensing capability with the aid of distance sensor.

Data Capabilities

Data transfer from the sensor to the arduino and from the arduino to the bluetooth device via the bluetooth module.

Interface Capabilities

Network interface of bluetooth module and inbuilt wifi module on arduino

Supporting Capabilities

Time synchronisation



Trustworthiness Viewpoints

Data Security

Ensuring all transmitted and stored data is secure from unauthorised access.

System Reliability

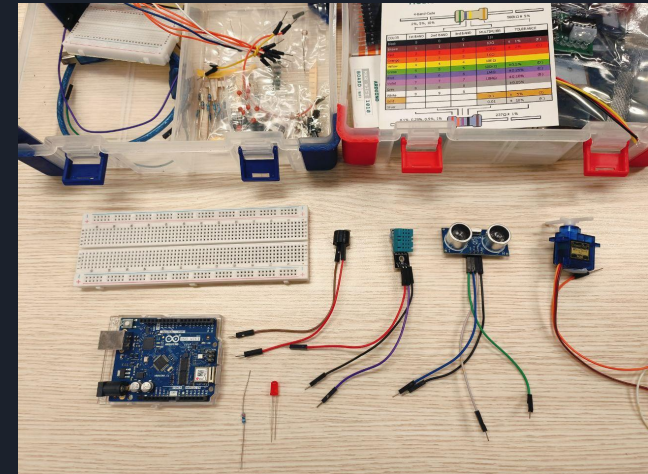
Must operate reliably under all expected conditions.

Privacy

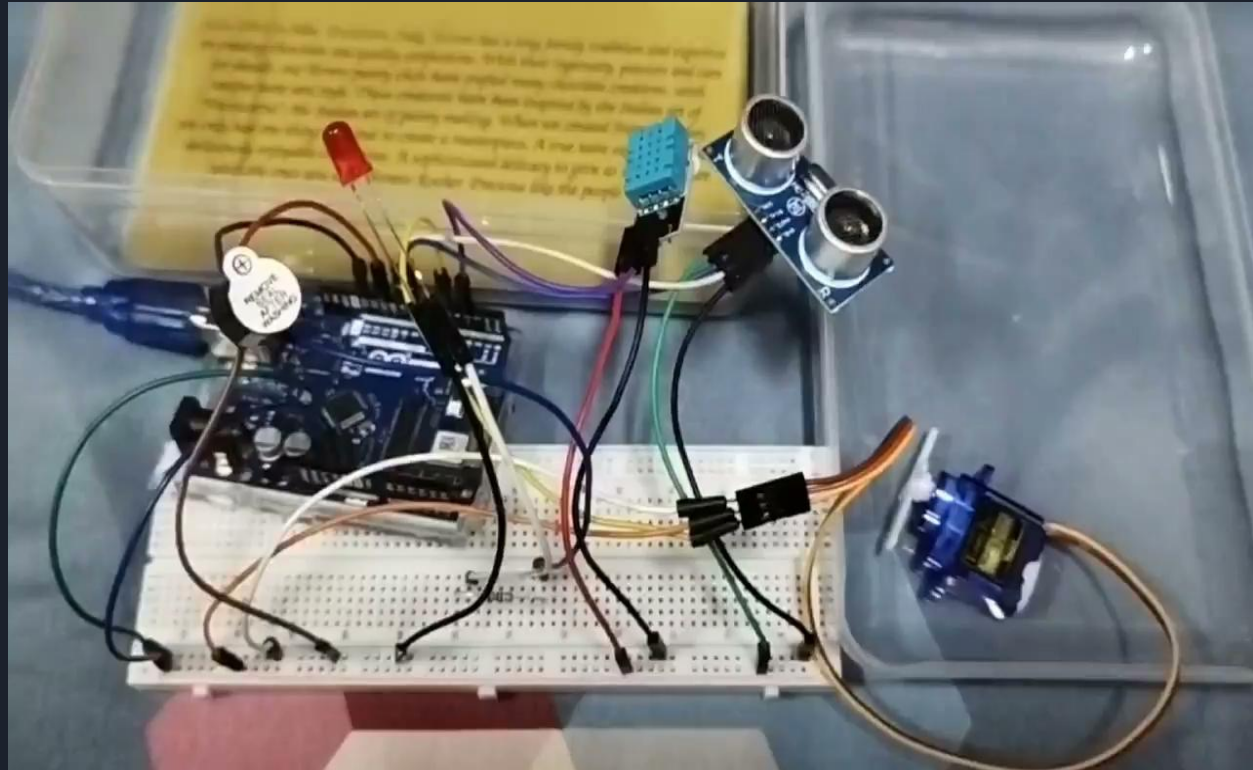
Protecting the privacy of the areas monitored by the sensors.

Purpose of the Equipment Used

- ★ Arduino Uno R3
 - Works as the mainboard of the system
- ★ Servo motor
 - Helps to rotate sonar sensor
- ★ Ultrasonic Sonar sensor
 - Is used to detect object by emitting ultrasonic wave
- ★ Temperature and Humidity sensor
 - Reads the current temperature of the environment to calculate exact velocity of the ultrasonic wave
- ★ Active buzzer and Red LED
 - Is used to turn on a warning sound and light when intrusion happens



Project Demonstration



Project Demonstration



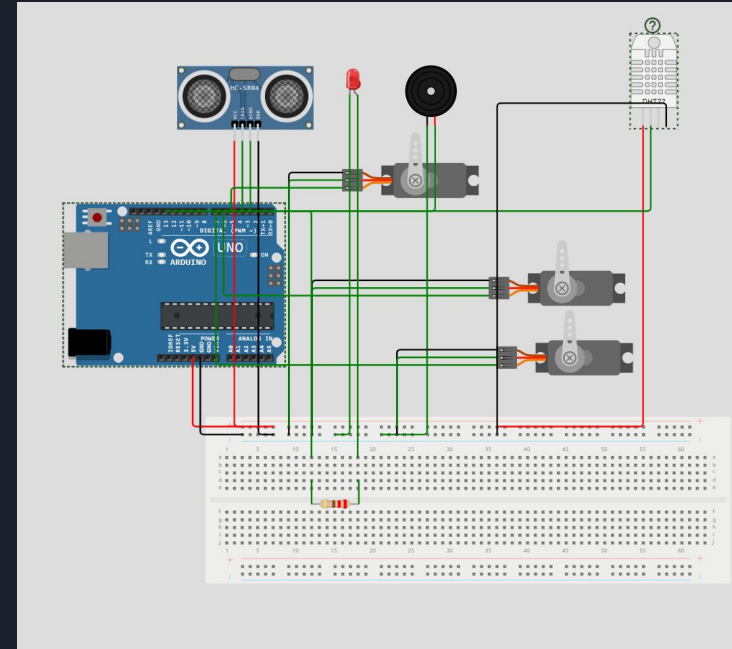
Arduino Simulation

Purpose:

The use of an Arduino simulator was to validate the project's design and functionality. The simulator allowed for testing of the code and circuit design in a virtual environment before implementation on physical hardware. This approach ensures that potential issues are identified and resolved early in the development process.

Project's Simulator Link:

<https://wokwi.com/projects/403831306271772673>





Data Transmission: 2 ways

- **Bluetooth Module**

- The Arduino detects an object within 30 cm and sends "Intruder!\n" via Bluetooth.
- The smartphone, paired and connected to the Arduino's Bluetooth module, receives this data through a dedicated app running in the background.
- The app reads the incoming data, recognizes the "Intruder!" message, and triggers a notification on the phone.

- **Google Sheet via Wifi**

- When the Arduino detects an intruder, it sends the data to the Google Apps Script endpoint.
- The script receives this data and appends it as a new row in the Google Sheet.
- The Google Sheet will generate the intrusion logs populating in real-time (or near real-time).



Limitations

- ★ Unable to get two servo motor acting as wheel to perform normally for the intruder detector robot movement.
- ★ Servo motor cannot rotate 360 degrees
- ★ Cannot detect objects moving at a high speed
- ★ Range of the sonar sensor is only 2cm to 400cm
- ★ Require power that are connected from Arduino to computer
- ★ GPS module must be positioned outdoors.

Limitations:

Observation and Findings

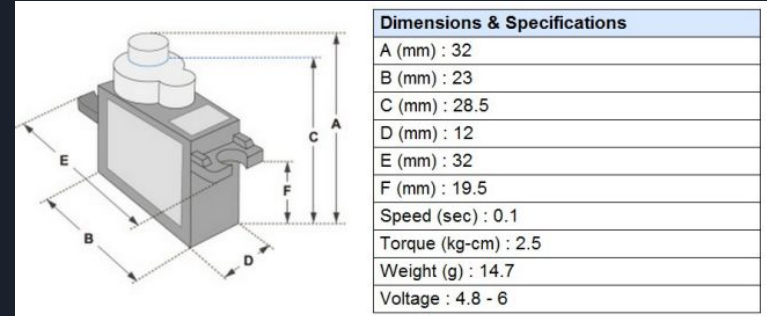
Observation:

Unable to run more than 1 servo smoothly.

- 3 servo motor: Only main servo run (abnormally), while 2 servo motor twitch.
- 2 servo motor: Servo1 move then pause then servo2 move then pause.
- 1 servo motor: Normal operation.

Finding:

1 servo motor take about 4.8 - 6 V



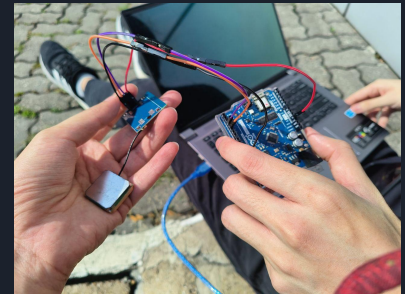
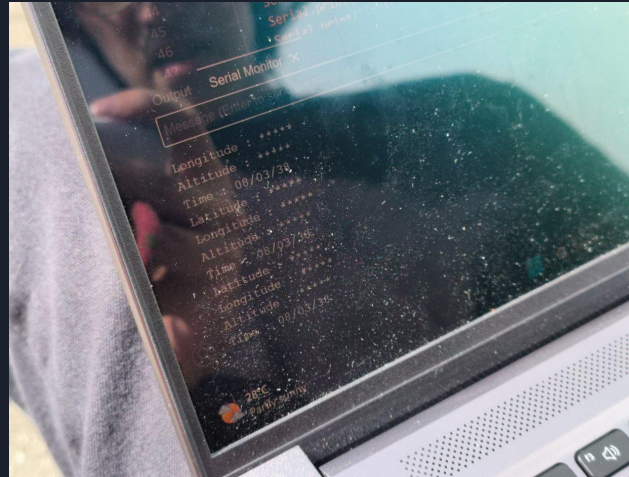
Limitations: Observation and Findings

Observation:

GPS not working

Finding:

GPS output requires the Arduino GPS module to be outside with a good view of the sky and has been there for a couple of minutes. (needs more time and stronger signals)





Evaluation of Characteristics

- Availability : Automation using sensors and to provide output to devices. However, may be limited due to lack of backup power supply, sensor failures.
- Composability & heterogeneity : Integrates multiple components into a cohesive system and handled by Arduino
- Functional management capability : Separating operational functions (like patrolling and threat detection) from management functions (such as configuration and updates) enhances security and reduces complexity.
- Network connectivity & communication : The system uses bluetooth for network connectivity, enabling monitoring. Network connectivity is essential for the system's operation and flexibility
- Real time capability : The system operates in real-time to respond to sensor inputs and control actuators. Real-time capability is crucial for timely responses to the surroundings changes.
- Context & content awareness : Measure its surroundings (Temperature, motion, movement), reacts to conditions by sending respective output signals to devices



Future improvements that can be made to the project/Concerns

01

Robust Encryption

Implement robust encryption for data transmission.

02

Regular Testing and Updates

Regular testing and updates to maintain system reliability and security.

03

Strict Access Controls

Strict access controls and privacy filters to ensure that monitoring is only for security purposes and respects privacy norms.

Learning points

Understanding IoT Integration

Key components used (sensors, Arduino, communication modules).
Importance of data security and real-time threat detection.

System Design and Functionality

System architecture and sensor integration. Data handling and alert mechanisms.



THANK you